Research Paper Assignment - Week 4

Applied AI I - Feature Importance & Ablation Study

Course: Angewandte KI I **Semester:** Winter 2025 **Assignment Type:** Research Paper **Weight:** 18% of Final Grade **Due Date:** Will be announced by the instructor and in moodle **Instructor:** Prof. Dr. Sigurd Schacht

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Assignment Overview

Research Question

"Which features matter and why? An ablation study on dataset X."

In this paper, you will conduct a systematic ablation study to identify which features contribute most to model performance on a real-world dataset. You will systematically remove features (individually and in groups) to understand their importance and interactions, comparing multiple feature selection methods to build an interpretable and efficient model.

Learning Objectives

By completing this assignment, you will:

1. Conduct systematic ablation experiments
2. Apply multiple feature selection techniques
3. Analyze feature importance and interactions
4. Design rigorous experimental protocols
5. Interpret and communicate feature contributions
6. Balance model performance with interpretability

Assignment Requirements

1. Dataset Selection

Requirements:

- **Source:** Reputable public source
  - UCI Machine Learning Repository
  - Kaggle Datasets
  - OpenML
  - Domain-specific repositories
  - Your own data (with proper description)
- **Characteristics:**
  - Minimum 1,000 samples
  - Minimum 8 features (more is better for ablation study)

  - Classification or regression task
  - Real-world dataset with meaningful features
  - Features should have interpretable meanings
- **Justification:**
  - Explain why this dataset is suitable for feature analysis
  - Describe what insights feature importance might reveal
  - Discuss domain context and why certain features might matter

Not Allowed: - Datasets with fewer than 8 features - Synthetic datasets without real-world context - Datasets where all features are equally important (e.g., pixel values) - Datasets already used in previous assignments —

2. Methodology

2.1 Data Preprocessing Document preprocessing steps: - Missing value treatment - Outlier handling - Feature scaling/normalization - Train/validation/test split (before any feature analysis!) - No feature engineering initially (analyze original features first)

2.2 Baseline Model

- Select ONE model architecture (e.g., Random Forest, Gradient Boosting)
- Train on ALL features
- Establish baseline performance
- Document model hyperparameters (keep constant throughout)

2.3 Feature Selection Methods Implement and compare at least **Two** of the following:

1. **Filter Methods:**
   - Correlation analysis
   - Mutual Information
   - Chi-square (classification)
   - ANOVA F-test
2. **Wrapper Methods:**
   - Recursive Feature Elimination (RFE)
   - Forward Selection
   - Backward Elimination
3. **Embedded Methods:**
   - Tree-based feature importance
   - Model coefficients
4. **Model-Agnostic Methods:**
   - Permutation Importance
   - SHAP values (optional, advanced)

2.4 Ablation Study Design **Required Experiments:**

1. **Individual Feature Ablation**
    - Remove each feature one at a time
    - Measure performance drop
    - Identify most critical features
2. **Cumulative Feature Addition**
    - Start with most important feature
    - Add features incrementally
    - Plot performance vs. number of features
3. **Feature Group Ablation**
    - Group features by type/category
    - Remove entire groups
    - Analyze group contributions
4. **Top-K Feature Selection**
    - Compare models with top 3, 5, 7, etc. features
    - Find optimal feature subset
    - Balance performance and complexity

2.5 Experimental Protocol

- **Cross-Validation:** Use k-fold CV (k=5 or 10)
- **Metrics:** Choose appropriate for your task
    - Classification: F1, ROC-AUC, Precision, Recall
    - Regression: R^2, RMSE, MAE
- **Reproducibility:** Random seeds, documented parameters

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3. Paper Structure (3-5 pages)

Abstract (150-250 words)

- Research question
- Dataset and domain
- Methods used
- Key findings (which features matter most)
- Practical implications

1. Introduction

- Motivation: Why feature selection matters for this problem
- Domain context and feature interpretability
- Research question clearly stated
- Preview of findings
- Paper organization

2. Related Work

- Overview of feature selection techniques

- Prior work on similar datasets/domains
- Ablation studies in literature
- How your work contributes
- Minimum 3-5 citations

3. Dataset and Preprocessing

- **Dataset Description:**
  - Source and citation
  - Domain and application
  - Number of samples and features
  - Feature descriptions (important!)
  - Target variable
  - Exploratory analysis
- **Feature Overview Table:**
  - Feature name, type, range, description
  - Domain meaning of each feature
- **Preprocessing:**
  - Missing value handling
  - Outlier treatment
  - Feature scaling
  - Train/test split strategy

4. Methodology

- **Model Selection:**
  - Chosen model and justification
  - Hyperparameters
  - Why suitable for this analysis
- **Feature Selection Methods:**
  - Description of each method used
  - Implementation details
  - Hyperparameters
- **Ablation Study Protocol:**
  - Individual feature removal
  - Cumulative addition
  - Group ablation
  - Top-K selection
- **Evaluation:**
  - Metrics and justification
  - Cross-validation strategy
  - Statistical testing approach

5. Experiments and Results **Required Results in Code or Paper:**

- **Table 1: Dataset Characteristics**
- **Table 2: Baseline Performance (All Features)**

- **Table 3: Feature Importance Rankings** (from different methods)
- **Table 4: Individual Feature Ablation Results**
- **Table 5: Top-K Feature Selection Performance**

Required Figures in Code or Paper:

- **Figure 1: Feature Correlation Matrix**
- **Figure 2: Feature Importance Comparison** (different methods)
- **Figure 3: Individual Feature Ablation** (performance drop when removed)
- **Figure 4: Cumulative Feature Addition Curve**
- **Figure 5: Performance vs. Number of Features**
- **Figure 6: Feature Group Contributions** (if applicable)

Analysis: - Which features are most important across methods? - Do different methods agree or disagree? - Performance vs. feature count trade-off - Interactions between features

6. Discussion Address:

- **Which features matter most?**
  - Consistent findings across methods
  - Surprising findings
  - Domain interpretation
- **Why do these features matter?**
  - Domain knowledge explanation
  - Feature interactions
  - Correlation vs. causation
- **Method Comparison:**
  - Which feature selection method worked best?
  - Agreement/disagreement between methods
  - Computational cost considerations
- **Practical Implications:**
  - Optimal feature subset recommendation
  - Performance vs. interpretability trade-off
  - Cost-benefit of data collection
  - Model deployment considerations
- **Feature Interactions:**
  - Do features work independently or together?
  - Redundant features
  - Complementary features
- **Limitations:**
  - Dataset limitations
  - Method limitations
  - Generalizability concerns

7. Conclusion

- Summary of key findings
- Direct answer: which features matter and why?
- Recommended feature subset
- Practical recommendations
- Future work

References

- Minimum 5 references
- Include key feature selection papers
- Dataset source
- Domain-specific work
- Consistent citation style (APA, IEEE, or ACM)

Evaluation Rubric

Total: 100 points

1. Dataset Selection & Preprocessing (15 points)

- Dataset appropriate for ablation study (5 pts)
- Features well-described and interpretable (5 pts)
- Preprocessing thorough and documented (5 pts)

2. Methodology & Experimental Design (25 points)

- Multiple feature selection methods implemented (8 pts)
- Ablation study properly designed and executed (10 pts)
- Cross-validation and evaluation appropriate (4 pts)
- Experimental protocol rigorous and reproducible (3 pts)

3. Results & Analysis (25 points)

- Complete ablation results reported (8 pts)
- Required visualizations clear and informative (8 pts)
- Feature importance comparisons thorough (5 pts)
- Statistical analysis conducted (4 pts)

4. Discussion & Interpretation (20 points)

- Features interpreted with domain knowledge (6 pts)
- Clear answer to "which features matter and why?" (5 pts)
- Method comparison insightful (4 pts)
- Limitations acknowledged (3 pts)
- Practical recommendations provided (2 pts)

5. **Writing Quality (10 points)**
    - Clear and concise academic writing (4 pts)
    - Proper structure followed (3 pts)
    - Figures/tables well-labeled and referenced (2 pts)
    - Grammar and formatting (1 pt)
6. **Technical Correctness (5 points)**
    - Code runs and is reproducible (2 pts)
    - Methods correctly implemented (3 pts)

BONUS POINTS (up to +10): - Advanced methods (SHAP, permutation importance) (+3) - Feature interaction analysis (+3) - Comprehensive sensitivity analysis (+2) - Exceptional visualizations (+2)

Formatting Requirements

Document Format

- **Length:** 4-5 pages (excluding references)
- **Font:** 11pt or 12pt, standard font
- **Spacing:** Single or 1.5 line spacing
- **Margins:** 2 cm all sides
- **Format:** PDF (required)

Code Submission

- Jupyter notebook (.ipynb)
- Well-commented and organized
- Reproducible (random seeds set)
- All experiments included
- requirements.txt or environment.yml

Tips for Success

Dataset Selection

Choose dataset where features have clear meanings Ensure enough features for meaningful ablation Consider domain where feature interpretation matters Look for features with different types/roles

Experimental Design

Keep model hyperparameters constant throughout Use same random seeds for fair comparison Apply feature selection AFTER train/test split Document all decisions and parameters Run experiments multiple times to check stability

Analysis

Look for consistent patterns across methods Interpret findings with domain knowledge Consider feature interactions, not just individual importance Visualize results clearly Check if top features make domain sense

Writing

Explain WHY features matter, not just WHICH Use domain knowledge to interpret patterns Discuss agreement/disagreement between methods Provide actionable recommendations Be honest about limitations

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Common Pitfalls to Avoid

Data Issues

x Applying feature selection before train/test split x Data leakage through preprocessing x Not documenting feature meanings x Choosing dataset with uninformative features

Methodology Issues

x Changing model hyperparameters between experiments x Using only one feature selection method x Not testing statistical significance x Ignoring feature interactions x Removing features without measuring impact

Analysis Issues

x Reporting importance scores without interpretation x Ignoring domain knowledge x Not explaining WHY features matter x Accepting results that don't make domain sense x Over-interpreting small differences

Writing Issues

x Not describing what features represent x Lists of numbers without interpretation x No practical recommendations x Missing domain context x Unclear which features to actually use

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Essential References

Feature Selection Methods

1. **Guyon, I., & Elisseeff, A. (2003).** "An introduction to variable and feature selection." *Journal of Machine Learning Research*, 3, 1157-1182.

2. **Chandrashekar, G., & Sahin, F. (2014).** "A survey on feature selection methods." *Computers & Electrical Engineering*, 40(1), 16-28.

Interpretability

3. **Molnar, C. (2020).** *Interpretable Machine Learning*. Available online.
4. **Ribeiro, M. T., Singh, S., & Guestrin, C. (2016).** "Why should I trust you?: Explaining the predictions of any classifier." *KDD*.

Example Research Questions

Good Examples:

"Which clinical features are most predictive of diabetes diagnosis, and can we reduce the number of required tests?"

"Do all 30 features in the credit card fraud dataset contribute to detection, or can we build an efficient model with fewer features?"

"Which housing characteristics most strongly influence price predictions, and how do they interact?"

Avoid:

x "Which features are important?" (too vague) x "Feature selection on dataset X" (not a research question) x "Comparing feature selection algorithms" (focus on features, not just methods)

Code Structure Template

```python
# ============================================
# Week 4: Feature Importance & Ablation Study
# Student: [Your Name]
# Dataset: [Dataset Name]
# ============================================

# -------- IMPORTS --------
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler
```

```python
from sklearn.ensemble import RandomForestClassifier  # or your chosen model
from sklearn.feature_selection import (
    mutual_info_classif,
    RFE,
    SelectKBest,
    f_classif
)
from sklearn.metrics import *

import warnings
warnings.filterwarnings('ignore')

# Set random seed
RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

# -------- CONFIGURATION --------
TEST_SIZE = 0.2
CV_FOLDS = 5
MODEL_PARAMS = {
    # Your model parameters
    'random_state': RANDOM_STATE
}

# -------- 1. LOAD DATA --------
# df = pd.read_csv('your_data.csv')

# -------- 2. EXPLORATORY ANALYSIS --------
# Feature descriptions
# Correlation analysis
# Distribution plots

# -------- 3. PREPROCESSING --------
# Handle missing values
# Treat outliers
# Feature scaling
# Train/test split (BEFORE feature selection!)

# -------- 4. BASELINE MODEL (ALL FEATURES) --------
# Train baseline with all features
# Evaluate performance
# Store baseline metrics

# -------- 5. FEATURE SELECTION METHODS --------

# 5.1 Filter Methods
```

```
# - Correlation
# - Mutual Information
# - Statistical tests

# 5.2 Wrapper Methods
# - RFE
# - Forward/Backward Selection

# 5.3 Embedded Methods
# - Model-based importance
# - L1 regularization

# -------- 6. ABLATION STUDY --------

# 6.1 Individual Feature Ablation
# For each feature:
#   - Remove feature
#   - Train model
#   - Measure performance drop
#   - Store results

# 6.2 Cumulative Feature Addition
# Start with empty set
# Add features one by one (by importance)
# Track performance

# 6.3 Top-K Feature Selection
# Try k = 3, 5, 7, 10, etc.
# Compare performance

# -------- 7. VISUALIZATIONS --------
# - Feature importance comparison
# - Ablation results
# - Performance vs. feature count
# - Correlation heatmap

# -------- 8. STATISTICAL ANALYSIS --------
# Compare feature subsets
# Significance testing

# -------- 9. RESULTS EXPORT --------
# Save figures
# Export results tables
```

Submission Guidelines

What to Submit

1. **Paper (PDF)**
   - Filename: `LastName_FirstName_Week4_Paper.pdf`
   - 3-5 pages, properly formatted
2. **Code (Jupyter Notebook)**
   - Filename: `LastName_FirstName_Week4_Notebook.ipynb`
   - Well-commented, reproducible
3. **Dataset**
   - If not publicly available, include data file
   - OR provide clear download instructions
4. **Dependencies**
   - `requirements.txt` or `environment.yml`

Submission Method

- Upload to Moodle
- **Deadline:** [To be announced]

Late Submission Policy

- 10% deduction per day late
- Maximum 3 days late accepted
- Contact instructor in advance for special circumstances

Academic Integrity

Collaboration Policy

- You must work in pairs
- Discuss approaches with classmates
- All code and writing must be your team's own
- Cite any external code sources
- Do not share code/results with other teams

Plagiarism

- All work must be original
- Properly cite all sources
- Paraphrasing without citation is plagiarism

FAQ

Q: How is this different from Paper 1? A: Paper 1 focused on comparing regularization methods. Paper 2 focuses on understanding which features in your dataset matter most and why.

Q: What if all features seem equally important? A: This is a valid finding if true, but carefully verify with multiple methods. Most real datasets have varying feature importance. Discuss why this might be the case.

Q: Do I need to use the same model as Paper 1? A: No, choose the model best suited for your dataset and task. But keep it constant throughout YOUR ablation study.

Q: How many features should I remove in the ablation study? A: Remove each feature individually at least once. Also try different Top-K subsets (e.g., top 3, 5, 7 features).

Q: Can I use SHAP or advanced interpretability methods? A: Yes! This can earn bonus points, but implement at least 3 standard methods first.

Q: What if feature selection methods disagree? A: Excellent observation! Discuss why different methods might rank features differently and which you trust more for your problem.

Q: Should I engineer new features? A: Focus on analyzing the original features first. You can briefly discuss potential engineered features in future work, but the main analysis should be on the dataset's original features.

Q: How do I handle feature groups? A: Group by domain meaning (e.g., all demographic features, all financial features). This depends on your specific dataset.

Checklist Before Submission

Paper Checklist

- ☐ Abstract clearly states which features matter
- ☐ All features in dataset described
- ☐ Multiple feature selection methods implemented
- ☐ Individual ablation results reported
- ☐ Cumulative addition analysis included
- ☐ All required figures and tables included
- ☐ Features interpreted with domain knowledge
- ☐ Clear answer to "which features matter and why?"
- ☐ Practical recommendations provided
- ☐ Minimum 5 references
- ☐ 3-5 pages (excluding references)
- ☐ PDF format

Code Checklist

- ☐ All imports at top
- ☐ Random seeds set
- ☐ Train/test split before feature selection
- ☐ All feature selection methods implemented
- ☐ Ablation experiments complete
- ☐ All visualizations generate correctly
- ☐ Code runs without errors
- ☐ Well-commented
- ☐ requirements.txt included

Submission Checklist

- ☐ Files named correctly
- ☐ All required files included
- ☐ Submitted before deadline
- ☐ Confirmation received

Key Differences from Paper 1

Since you've already completed Paper 1, here's what's different:

Paper 1 (Regularization): - Compared different modeling approaches - Focus: Does regularization help? - Methods were the focus

Paper 2 (Feature Analysis): - Keep model constant - Focus: Which data features matter? - Features are the focus - More emphasis on domain interpretation - Ablation study is the key methodology

Expectations: - Less hand-holding in instructions - More emphasis on interpretation and domain knowledge - Higher expectations for experimental rigor - Stronger focus on practical insights

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Learning Outcomes

After completing this assignment, you should be able to:

Design systematic ablation studies **Apply** multiple feature selection techniques **Interpret** feature importance with domain knowledge **Analyze** feature interactions and redundancy **Communicate** which features matter and why **Recommend** optimal feature subsets for deployment **Balance** model performance with interpretability

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Good Luck!

Remember: The goal is not just to identify important features, but to understand WHY they matter in your domain context. The best papers will combine rigorous experimentation with thoughtful domain interpretation.

Focus on: - Thorough experimental design - Clear, interpretable results - Domain-grounded explanations - Practical, actionable insights

This is your opportunity to deeply understand what makes your dataset tick. Enjoy the exploration!